

Project Management

by PUL079BEI007

by PUL079BEI008

by PUL079BEI014

April 5, 2026

Notes

- Our subject code is CT 658 and CT 701.
- This document incorporates for both, electronics and computer department.
- If a question was asked in Regular Exam, it is kept in boldened form. If it was asked in back exam, it is in regular font.
- Months are marked as:
 - Ba: Baisakh (81)
 - Jth: Jestha
 - Asa: Ashar
 - Shr: Shrawan (70, 71, 72, 73)
 - Bh: Bhadra
 - Ash: Ashwin (75)
 - Ka: Kartik
 - Mng: Mangsir
 - Po: Poush
 - Ma: Magh

– Ch: Chaitra (69, 70, 71, 72, 74)

Contents

1	Introduction (2 hours)	4
1.1	Definition of project and project management	4
1.2	Project objectives	4
1.3	Classification of projects	4
1.4	Project life cycle	4
2	Project Management Body of Knowledge (4 hours)	4
2.1	Understanding of project environment	4
2.2	General management skill, effective and ineffective project managers	5
2.3	Essential interpersonal and managerial skills	5
2.4	Energized and initiator, communication, influencing, leadership, motivator, negotiation, problem solver, perspective nature, result oriented, global illiteracies, problem solving using problem trees	5
3	Portfolio and Project Management Institutes' (PMI) Framework (2 hours)	6
3.1	Portfolio	6
3.2	Project management office	6
3.3	Drivers of project success	6
3.4	Inhibitors of project success	6
4	Project Management (4 hours)	6
4.1	Advantages of project management	6
4.2	Project management context as per PMI	6
4.3	Characteristics of project life cycles, representative project life cycles, IT Product Development Life Cycle, Product Life Cycle and Project Life Cycle	6
4.4	System Development methodologies, role and responsibilities of key project members	7
5	Project and Organizational structure (2 hours)	7
5.1	System view of project management	7
5.2	Functional organization, matrix organization	7
5.3	Organizational structure influences on projects	7
6	Project Management Process Groups (2 hours)	8
6.1	Project management processes	8
6.2	Overlaps of process groups in a phase, mapping of project management process groups to area of knowledge	8
7	Project Integration Management (4 hours)	8
7.1	Develop project charters	8
7.2	Develop preliminary project scope statement, Develop project management plan, Direct and manage project execution, monitor and control project work	8
7.3	Integrated change control, close project	8
7.4	Project scope management, Create Work Break Down Structure, Scope verification, Scope control	9
8	Project Time Management (4 hours)	9
8.1	Activity definition, decomposition of activities, activity attributes	9
8.2	Activity sequencing, precedence relationship, network diagram, precedence diagram method, arrow diagramming method	9

8.3	Activity resources estimating, determining resource requirements	9
8.4	Schedule development and control, principles of scheduling, milestones, forward pass, backward pass, critical path method, critical chain technique, gantt chart, schedule control . . .	9
9	Project Cost Management (4 hours)	10
9.1	Cost and project, cost management	10
9.2	Cost estimating, types of cost estimates, estimating process and accuracy	10
9.3	Cost budgeting, cost aggregation, deriving budget from activity cost	10
9.4	Cost control process, cost control methods, earned value management, EVM benefits, variance analysis	10
10	Project quality management (3 hours)	11
10.1	Quality theories	11
10.2	Quality planning, project quality requirements, cost of quality, quality management plan .	11
10.3	Quality assurance, quality audit, approach to a quality audit	11
10.4	Quality control process, control chart, pareto charts, testing of IT system, the test life cycle	11
11	Project Communication Management (3 hours)	12
11.1	Importance of communication management	12
11.2	Communications planning process, communication requirement analysis, organizing and conducting effective meeting	12
11.3	Information distribution process	12
11.4	Performance reporting process, integrated reporting system	12
12	Project Risk Management (4 hours)	13
12.1	Understanding Risk, project risk	13
12.2	Risk management planning process, risk management plan	13
12.3	Risk identification, risk identification techniques	13
12.4	Qualitative risk analysis process	13
12.5	Quantitative risk analysis process, modeling techniques	13
12.6	Risk response planning, resolution of risk, strategies for negative risks or threats, strategies for positive risks or opportunities	13
12.7	Risk monitoring and control process	13
13	Project Procurement Management (3 hours)	14
13.1	Procurement management process flow	14
13.2	Plan purchases and acquisition process	14
13.3	Plan contracting process, standard forms, evaluation criteria	14
13.4	Request seller response process	14
13.5	Select seller process	14
13.6	Contract administration process	14
13.7	Contract closure process	14
14	Developing Custom Processes for IT projects (3 hours)	15
14.1	Developing IT project management methodology	15
14.2	Moving forward with customized management processes	15
14.3	Certified associate in project management	15
14.4	Project management maturity	15
14.5	Promoting project Excellency through awards and assessment	15
14.6	Certification process flow	15
14.7	Code of ethics	15

14.8 Future trends	15
15 Balanced scorecard and ICT project management (1 hour)	15
16 Other Short Notes and Miscellaneous	16

1 Introduction (2 hours)

1.1 Definition of project and project management

1. Define project management. Why it is important? And how do you identify the completion of project phases? [1.5+2+1.5] (81 Ch)
2. Define project. Differentiate between project and operation. [2+2] (79 Ch)
3. How can you relate the characteristics of project with your minor project? [4] (78 Ch)
4. What are the triple constraints of project? How is project different from operational work? [3+2] (75 Ashwin)
5. What is the project? Explain the triple constraints of project with figure and describe their relationship. [1+4] (74 Ashwin)
6. What is project? Explain the triple constraints of project with figure. [2+3] (72 Ch)
7. What is a project, and what are its main attributes? How is a project different from what most people do in their day-to-day jobs? Discuss the specific attributes that are specific to IT related projects. [4] (69 Ch)
8. List out the characteristics of a project. Explain the role of effective feasibility study for the successful completion of a project. [2+2] (70 Ashad)
9. What is a project? List out its characteristics. (70 Ashad)

1.2 Project objectives

1.3 Classification of projects

1. How can IT projects be classified? Write about the challenges of IT projects. [2+2] (73 Shrawan)

1.4 Project life cycle

1. What do you mean by Project Development Life Cycle? Explain various phases of the project development life cycle. [5] (74 Ch)
2. Explain the characteristics of project life cycle with diagram. [2+3] (74 Ashwin)
3. Explain the characteristics of project life cycle with diagram. [2+3] (73 Shrawan)
4. What are the phases in project life cycle? How does a project life cycle differ from a product life cycle? [5] (70 Ashad)

2 Project Management Body of Knowledge (4 hours)

2.1 Understanding of project environment

1. Define PMBOK. How the external environmental influences on ICT project? [2+3] (79 Ch, 78 Ch)
2. What is PMBOK? And how the external environmental influence on ICT project? [2+3] (75 Ashwin)
3. What is PMBOK? And how the external environmental influences on ICT project? [2+3] (74 Ch)

2.2 General management skill, effective and ineffective project managers

1. What do you mean by general management skill? What are the essential interpersonal and managerial skills? [2+3] (81 Ch)
2. One of the famous actors, Bill Cosby, once said “I don’t know the key to success but the key to failure is trying to please everybody.” How relevant is this statement in the field of IT System Project Management? What are other traits of successful Project Manager? Explain. [6] (80 Ch)
3. Briefly explain the traits of being an effective and ineffective project manager? [2+2] (71 Ch)
4. Explain different general management skills necessary to be a good project manager. [5] (69 Ch)

2.3 Essential interpersonal and managerial skills

1. What are the essential managerial skill and soft skill to be a successful Project Manager? [3+2] (75 Ashwin)
2. What are the essential interpersonal and managerial skills to be a successful project manager?[2+3] (74 Ch)
3. What are the managerial skills and essential soft skills to be a successful project manager? [2+3] (74 Ashwin)
4. What are the essential interpersonal and managerial skills to be a successful project manager? [5] (72 Ch)
5. What are different skill sets required by a project manager? Briefly explain each of them.[5] (70 Ashad)
6. What is the role of project manager? What are suggested skills for all project managers and for information technology project managers? [5] (70 Ashad)

2.4 Energized and initiator, communication, influencing, leadership, motivator, negotiation, problem solver, perspective nature, result oriented, global illiteracies, problem solving using problem trees

1. What is project communication management? Why it is important to complete project successfully? Find the no. of communication channels among 7 people. [1+2+2] (81 Ch)
2. Why communication management is important in IT project? As a project manager how do you conduct meeting effectively? [3+3] (79 Ch)
3. Why communication management is important in IT project? As project manager, how do you write effective email? [2+2] (78 Ch)
4. Compare Formal versus informal communication in PM. [4×3] (80 Ch)

3 Portfolio and Project Management Institutes' (PMI) Framework (2 hours)

3.1 Portfolio

1. Define PMI framework. What are the key benefits of project portfolio management? [2+3] (75 Ashwin)
2. What are the key benefits of project portfolio management? [2] (75 Ashwin)
3. What are the key benefits of project portfolio management? Explain the characteristics of project life cycle with diagram. [2+3] (73 Shrawan)
4. What are the key benefits of project portfolio management? [2] (74 Ashwin)
5. Write short notes on: Portfolio Management [3] (81 Ch)

3.2 Project management office

3.3 Drivers of project success

3.4 Inhibitors of project success

1. Explain with example the concept of drivers of project success and inhibitors of project success. [2+3] (70 Ashad)
2. What are the major causes of failure of the ICT project? Describe what bodies of knowledge are required by a PM to contribute for a successful project implementation. [3+4] (72 Ch)
3. Discuss briefly the challenges in IT project in Nepal. [4] (80 Ch)
4. Define project portfolio management. What are the reasons for failure of the ICT projects? [2+5] (78 Ch)
5. What are the major causes of failure of the ICT project? Explain how organizational structures influence projects? [2+3] (74 Ashwin)

4 Project Management (4 hours)

4.1 Advantages of project management

4.2 Project management context as per PMI

1. Define PM context as per PMI. Explain the role and responsibility of key project member. [1+4] (74 Ashwin)

4.3 Characteristics of project life cycles, representative project life cycles, IT Product Development Life Cycle, Product Life Cycle and Project Life Cycle

1. What are the differentiation between product life cycle & project life cycle? Describe different types of organizational structure. [2+4] (81 Ch)
2. Explain the characteristics of project life cycle with diagram. [2+3] (74 Ashwin)

4.4 System Development methodologies, role and responsibilities of key project members

1. What is a Software Development Life Cycle (SDLC)? Explain any one of its kind that you prefer in developing an IT project. Why? [2+5] (69 Ch)
2. Discuss about IT project management methodology. [5] (70 Ashad)
3. What is a Maturity Model for software development? Explain them. [5] (70 Ashad)
4. Describe project management? Explain the roles and responsibilities of key project members. [2+2] (71 Ch)
5. Explain role and responsibilities of key project members. [6] (78 Ch)

5 Project and Organizational structure (2 hours)

5.1 System view of project management

1. What is system view of project management? Explain about three sphere model of system management. [2+4] (81 Ch)
2. What is system view of project management? Explain about three sphere model of system management. [6] (80 Ch)
3. What is system view of project management? Explain about three sphere Model of system management. [2+3] (79 Ch)
4. What is system view of project management? Explain, how organizational structures influence the IT projects? [2+3] (78 Ch)
5. What is system view of project management? Explain about three sphere model of system management. [2+3] (75 Ashwin)

5.2 Functional organization, matrix organization

1. Define organization structure. What are the different organizational structures and which type of structure do you feel is the most effective in ICT project? [1+4] (75 Ashwin)
2. Differentiate and compare among various organizational structures, with examples. [6] (80 Ch)
3. Describe about functional organization structure. What are the disadvantages of projectized organizational structure? [3+2] (79 Ch)
4. Most of the project follows functional organizational structure. If you agree, justify. [4] (69 Ch)
5. Explain a Matrix Organization Structure with its advantages and disadvantages. [4] (70 Ashad)

5.3 Organizational structure influences on projects

1. Explain how organizational structures influence projects? [2+3] (74 Ashwin)

6 Project Management Process Groups (2 hours)

6.1 Project management processes

1. Describe about project process groups. (79 Ch)
2. Write the concept of project management process groups. How is it related to project management knowledge areas? [3+2] (75 Ashwin)
3. Discuss about Project Management Process Groups. [4] (74 Ch)
4. Discuss the concept of project management process groups (PGs). How is it related to project management knowledge area? Give the example of two processes with necessary inputs, tools and techniques and outputs. [4] (70 Ashad)

6.2 Overlaps of process groups in a phase, mapping of project management process groups to area of knowledge

1. Explain overlaps of project process groups with clear diagram. [5] (78 Ch)

7 Project Integration Management (4 hours)

7.1 Develop project charters

1. What is a project charter? How do you develop a project charter, explain the inputs and tools and techniques to develop it. [2+5] (71 Ch)
2. What is project charter? What type of information should be included while making the project charter? Explain with example. [1+4] (81 Ch)
3. What is Project Integration Management? What type of information should include while making the project charter? [2+3] (79 Ch)
4. Define the Project Integration Management. What are the essential information required to create Project Charter? [3+3] (78 Ch)
5. Define the Project Integration Management and what is the essential information required to create Project Charter? [2+3] (75 Ashwin)
6. Define project integration management. What type of information should include while making the project charter? [2+3] (74 Ashwin)

7.2 Develop preliminary project scope statement, Develop project management plan, Direct and manage project execution, monitor and control project work

7.3 Integrated change control, close project

1. What are the key features in change control on IT projects? Explain about input, tools and techniques and output of integrated change control. [3+3] (73 Shrawan)
2. Explain about the integrated change control in detail. [6] (72 Ch)

3. Explain the integrated change control process in depth. [4] (70 Ashad)
4. Explain about Integrated Change Control in IT project development. [5] (70 Ashad)

7.4 Project scope management, Create Work Break Down Structure, Scope verification, Scope control

1. What is project scope management? And compare Project scope with product Scope. [2+2] (73 Shrawan)
2. What are the essential components of project scope management? Explain. [5] (71 Ch)
3. Write short notes on: WBS [3] (81 Ch)
4. Define work break down structure and its importance in project management. What are different ways/approaches to prepare a work breakdown structure for a project? [5] (70 Ashad)
5. Define WBS technique in scope management. [3] (70 Ashad)
6. Discuss the process of defining project scope in more detail as a project progresses, going from information in a project charter to a project scope statement, WBS and WBS dictionary. [5] (69 Ch)
7. Imagine you have got the role of Project Manager for a software project that will make personalized and secure access of your semester-end mark-sheet possible. This will relieve Exam Control Office for printing semester end marksheet rather students will browse and download semester end mark-sheet as pdf only through either using browser or ECD app. (80 Ch)
 - a) Prepare a brief project scope document for the above system project work. [7]
 - b) While making the process groups, what are the important considerations that you have to manage and take care of? [5]

8 Project Time Management (4 hours)

8.1 Activity definition, decomposition of activities, activity attributes

8.2 Activity sequencing, precedence relationship, network diagram, precedence diagram method, arrow diagramming method

1. Why is it important to determine activity sequencing on projects? What are different diagrams/methods that can be used to sequence activities in the project? [5] (70 Ashad)
2. Write short notes on: Arrow Diagramming Method. (69 Ch)

8.3 Activity resources estimating, determining resource requirements

8.4 Schedule development and control, principles of scheduling, milestones, forward pass, backward pass, critical path method, critical chain technique, gantt chart, schedule control

1. Consider the following project and answer the followings:[2+2+1+2+2] (81 Ch) [Table given in original]

2. KasthaPur, an innovative company list out the following activities with its duration in week for upcoming project. [10] (80 Ch) [Table given]
3. Construct a network diagram for this project and find the critical path using data below. Calculate the Late Start, Late Finish and find the total float. [3+1+1+2] (79 Ch) [Table]
4. Consider the following table of activities. (78 Ch) [Table]
5. Construct a network diagram for this project and explain what is meant by critical path? [1+2+2+1] (75 Ashwin) [Table]
6. Consider the following project and answer the followings: [2.5x4] (74 Ch) [Table]
7. Draw the network diagram and find the critical path using data below. Calculate the forward pass and backward pass and find the total float time. [2+2+2] (74 Ashwin) [Table]
8. Produce a critical path network diagram, showing the earliest start times and latest start times for each task, using the data in the table below. Calculate the total project time and total network slack time. [5+3+2] (73 Shrawan) [Table]
9. A project work consists of the following activities as listed below in table. [9] (71 Ch) [Table]
10. Write short notes on: Critical Chain Scheduling. (70 Ashad)
11. Project Crashing is termed as optimal point of trade-off between project cost and project time for any project while assessing delayed or over-run project. Do you agree or not? In any case justify with the help of right-side graph. [6] (80 Ch)

9 Project Cost Management (4 hours)

9.1 Cost and project, cost management

9.2 Cost estimating, types of cost estimates, estimating process and accuracy

9.3 Cost budgeting, cost aggregation, deriving budget from activity cost

9.4 Cost control process, cost control methods, earned value management, EVM benefits, variance analysis

1. Define EVM. Your project is scheduled for 2 years. Nine months into the project, while the total project budget is Rs 42,00,000. You have already spent Rs 16,50,000. CPI is 0.875. Calculate EV, EAC, ETC, VAC and is the project is behind the schedule or ahead schedule? [2+1+1+1+1+1+2] (81 Ch)
2. Define EVM. Suppose you are the project manager on an IOT project. The project is set to be completed in 10 months with an estimated cost of Rs 500,000. The project has been running for 5 months now, the team has spent Rs 220,000 and completed an amount of work worth Rs 255,000. Calculate CPI and SPI and Share your conclusion. [1+2+2+1] (79 Ch)
3. What is EVM? Calculate the schedule and cost variance for a project whose estimated budget is Rs 100,000 and estimated time is 50 man months. 5 people were involved in the project. After 7 months, 60% of work was completed and Rs 80,000 was expended. Also estimate the new cost and time for the project. [2+5] (78 Ch)
4. Compare Project Cost management versus Earned Value management. [4x3] (80 Ch)

5. Suppose you have IT project, which might look after project planning: [2.5+2.5+1] (75 Ashwin) [Table]
6. What is EVM? Suppose you are three month into a six month project. Assume that the budget burn rate is constant and the Budget at Completion (BAC) is Rs 1,20,000/- and Actual cost is Rs 65,000/- and Schedule Performance Index is 1.2. Find the CPI of the project and Estimate at Completion (EAC). Is the project is over budget or under budget? [1+2+1+1] (74 Ch)
7. What is EVM? The project has been planned that total estimated cost of project is Rs 5 Lakhs and there are 20 widgets to complete in 10 months duration. At 5th months, it reported that project was completed 40% and it has been spent Rs. 3 Lakhs only. Now calculate Cost Variance and Schedule Variance. Is the project is behind the schedule or ahead schedule? [1+2+2+1] (74 Ashwin)
8. What is earned value analysis? A project is scheduled for the time of 12 months. The estimated cost of project is 400000. After 3 months, evaluation is done and it is identified that 40% of work is accomplished but 200000 cost has been incurred. Now calculate cost and schedule variance for the project. [2+6] (73 Shrawan)
9. Suppose you are managing a software development project. The project is expected to be completed in 8 months at a cost of Rs.50,000/- per month. After 2 months, you realize that the project is 30 percent completed at a cost of Rs 200,000/-. Determine whether the project is on-time and on-budget after 2 months. Calculate Cost and Schedule Performance Index. [8] (72 Ch)
10. If schedule performance index (SPI) is 0.75 in a mega project undergoing near Devikapur district with earned value of being 60. Now calculate the planned value and also state whether the project is ahead schedule or behind schedule. [6] (70 Ashad)
11. If earned value is twice its actual cost for a project, calculate its cost performance index and cost variance percentage. Is the project over/under budget? [6] (71 Ch)
12. Given the following information for one-year project, use Earned Value Management (EVM) method to calculate, cost variance, schedule variance, cost performance index (CPI) and Schedule performance index (SPI) for the project. [6] (70 Ashad)

10 Project quality management (3 hours)

10.1 Quality theories

10.2 Quality planning, project quality requirements, cost of quality, quality management plan

10.3 Quality assurance, quality audit, approach to a quality audit

10.4 Quality control process, control chart, pareto charts, testing of IT system, the test life cycle

1. Quality is one of the most important factors to be controlled for effective delivery of project objectives. How quality assurance and quality control are implemented in order to deliver a successful project? [2+3] (78 Ch)
2. Quality is one of the most important factors to be considered for effective delivery of project objectives. How quality assurance and quality control are implemented in order to deliver a successful project? [5] (75 Ashwin)

3. Quality is one of the most important factors to be controlled for effective delivery of project objectives. How quality assurance and quality control are implemented in order to deliver a successful project? [5] (74 Ashwin)
4. Quality is one of the most important factors to be controlled for effective delivery of project objectives. How quality assurance and quality control are implemented in order to deliver a successful project? Describe. [8] (72 Ch)
5. What are tools and techniques for Total Quality Management? And write possible steps to improve quality IT project. [4+3] (73 Shrawan)
6. What do you understand by Quality planning, Quality Assurance and Quality Control? Explain different approaches to these processes. [4] (70 Ashad)
7. What do you understand by Quality in the context of project management? Discuss quality control process and its major outputs. [5] (69 Ch)
8. Is there always a tradeoff between quality and productivity? Explain with an IT related example. [3] (71 Ch)
9. What are the possible steps to improve project quality? [4] (71 Ch)
10. Write short notes on: Six sigma. [5x4] (71 Ch)
11. Write short notes on: Control charts vs. Cause and effect charts for quality assessment. [3x3] (73 Shrawan)
12. Write short notes on: Cause and effect diagram. [2.5x4] (74 Ch)
13. How Pareto charts help to achieve better quality project? [5] (74 Ch)
14. Consider below data about customer complaint on e-commerce portal and their services. Draw a free-hand sketch of Pareto diagram after having Pareto analysis with clearly showing all necessary calculations. As per Pareto principles, which are the first three complaints should be addressed first? What will be the coverage of issues by these solving such three issues? [6+2] (80 Ch) [Table]

11 Project Communication Management (3 hours)

11.1 Importance of communication management

11.2 Communications planning process, communication requirement analysis, organizing and conducting effective meeting

11.3 Information distribution process

11.4 Performance reporting process, integrated reporting system

1. A project manager is having trouble getting a project member to complete their tasks as assigned. What type of communication would the project manager want to use to address this problem initially and why? [5] (74 Ch)
2. Why communication management is important in IT project? How to run the effective meeting in project? [3+2] (74 Ashwin)

3. What is difference between communication skills and communication management? How does the communication skill help to resolve conflicts in ICT project? Explain with example. [2+4] (72 Ch)
4. Why better communication management is critical for projects? Discuss the communication management plan that should be considered for ICT projects. [5] (70 Ashad)
5. Why better communication is critical for ICT projects? Discuss the contents of communication management plan that should be considered. [5] (69 Ch)
6. Write short notes on: Horizontal vs. vertical communication and their degree of formality. [3×3] (73 Shrawan)
7. Why documentation and reporting system is required in a project? Explain the hazards of communication error in project? [2+3] (75 Ashwin)
8. Why reporting system and documentation are required in a project? Explain the hazard of communication errors in a project. [2+3] (74 Ashwin)
9. Explain various tools and techniques for performance reporting. [5] (71 Ch)
10. Explain about the necessity of information distribution and its tools and techniques. [5] (70 Ashad)
11. Write short notes on: Project management information system. (70 Ashad)

12 Project Risk Management (4 hours)

12.1 Understanding Risk, project risk

12.2 Risk management planning process, risk management plan

12.3 Risk identification, risk identification techniques

12.4 Qualitative risk analysis process

12.5 Quantitative risk analysis process, modeling techniques

12.6 Risk response planning, resolution of risk, strategies for negative risks or threats, strategies for positive risks or opportunities

12.7 Risk monitoring and control process

1. What is project risk management? What are the risk identification techniques? [2+3] (81 Ch)
2. Explain about risk management plan. What are the different sources of risk in IT project? [2+3] (79 Ch)
3. What is project risk? Explain about risk management plan. [1+4] (78 Ch)
4. Discuss briefly on risk identification techniques. Create a sample risk register of any ICT related project. [2+3] (80 Ch)

5. Write short notes on: Quantitative Risk Analysis. [3] (81 Ch)
6. Write short notes on: Sensitivity analysis. [5x4] (71 Ch)
7. Write short notes on: Decision tree analysis. [5x4] (71 Ch)
8. Why risk response planning is important in project? What are the response strategies for negative risk? [2+3] (75 Ashwin)
9. Why risk response planning is important in project? What are the response strategies for negative risk? [2+3] (74 Ch)
10. Risk management is an essential part of project management. Describe the risk identification techniques in IT project. [5] (74 Ashwin)
11. Risk Management is an essential part of project management. Describe three typical risks that can occur in a software project and for each of these risks suggest two possible countermeasures. [8] (73 Shrawan)
12. What are different tools and techniques for risk identification? Discuss brainstorming and Delphi Technique for risk management. [4] (70 Ashad)
13. Write short notes on: Delphi Technique. [2x4] (75 Ashwin)
14. Write short notes on: Categories of Risk. (70 Ashad)
15. Write short notes on: Decision Tree vs. Tornado Analysis for risk management. [3x3] (72 Ch)

13 Project Procurement Management (3 hours)

13.1 Procurement management process flow

13.2 Plan purchases and acquisition process

13.3 Plan contracting process, standard forms, evaluation criteria

13.4 Request seller response process

13.5 Select seller process

13.6 Contract administration process

13.7 Contract closure process

1. What do you mean by Project Procurement Management? Explain the different processes adopted in the procurement management? [2+4] (81 Ch)
2. What do you mean by Project Procurement Management? What are the different processes adopted for procurement in ICT project? [2+3] (80 Ch)
3. Explain about procurement management process flow. [2+3] (79 Ch)
4. What do you mean by Project Procurement Management? Explain the different process adopted for procurement in ICT project? [1+4] (75 Ashwin)

5. What do you mean by Project Procurement Management? Explain the different process adopted for procurement in ICT project? [1+4] (74 Ch)
6. What do you mean by Project Procurement Management? Explain the different process adopted for procurement in ICT project? [1+4] (74 Ashwin)
7. What is a procurement process? How is it performed in a project? [1+4] (70 Ashad)
8. What do you mean by Project Procurement management and what are the different processes adopted for procurement? [5] (71 Ch)
9. Write short notes on: Quotation based purchase vs. Tender based purchase for procurement process. [3×3] (72 Ch)
10. Write short notes on: Contract Closure Procedure. [2×4] (75 Ashwin)
11. Write short notes on: Contract closure procedure. [2×4] (74 Ashwin)
12. Write short notes on: Contract Closure Procedure. (73 Shrawan)

14 Developing Custom Processes for IT projects (3 hours)

14.1 Developing IT project management methodology

14.2 Moving forward with customized management processes

14.3 Certified associate in project management

14.4 Project management maturity

14.5 Promoting project Excellency through awards and assessment

14.6 Certification process flow

14.7 Code of ethics

14.8 Future trends

1. Write short notes on: Need custom processes for IT project [4] (79 Ch) [2] (78 Ch)
2. Write short notes on: Future trend of IT projects [4] (79 Ch)
3. Write short notes on: Project management maturity model. [3×3] (72 Ch)
4. Write short notes on: Project Management Maturity. [5×4] (71 Ch)
5. Write short notes on: ICT Code of Ethics [2×4] (75 Ashwin) [2×4] (74 Ashwin)
6. Write short notes on: Trends in cloud computing [4×5] (70 Ashad)

15 Balanced scorecard and ICT project management (1 hour)

1. Write short notes on: Balanced Scorecard (BSC) [3] (81 Ch) [2] (78 Ch)[2×4] [2.5×4] (74 Ch)[2×4] (74 Ashwin) (73 Shrawan) [3×3] (72 Ch) [4×5] (70 Ashad) [5×4] (71 Ch) (69 Ch)
2. Compare KPI versus Balanced scorecard [4] (80 Ch)

16 Other Short Notes and Miscellaneous

1. Write short notes on: Statement of Work (SOW) [3] (81 Ch)
2. Write short notes on: Project Triple Constraints [3] (81 Ch) [4] (79 Ch)
3. Compare Project Manager versus Program Manager as per PMI [4] (80 Ch)
4. Write short notes on: Break-even Point [2.5×4] (74 Ch)
5. Write short notes on: Traceability vs. Adaptability in reviewing steps [3×3] (73 Shrawan)
6. Write short notes on: Consistency vs. completeness in requirements engineering [3×3] (72 Ch)
7. Write short notes on: Responsibility assignment matrix [3×3] (72 Ch)
8. Write short notes on: Expert Judgement (69 Ch)
9. Write short notes on: Project stakeholders (70 Ashad)
10. Write short notes on: COCOMO for IT project. [2.5×4] (74 Ch)
11. Write short notes on: Constructive Cost Model (COCOMO). (70 Ashad)
12. The clothing manufacturer, Neplai Luga, is considering introduction a line of cargo pants made entirely from hemp. The project costs NRs. 4.6 million and will generate cash flows of NRs. 1 million for 5 years. What is the payback period? If the interest rate is 0.3% per month, what is the project's NPV? Should the project be accepted? Why or why not? [10] (74 Ch)
13. Write short notes on: SWOT Analysis. [2×4] (75 Ashwin) (73 Shrawan)
14. Explain the key elements of project proposal. [5] (73 Shrawan)
15. Being an IT project manager how are you going to manage an IT based project that demands regular updates with new trends in market. [5] (70 Ashad)
16. Discuss the challenges of outsourcing. [5] (69 Ch)
17. Why are organization moving towards the trend of outsourcing? Discuss the challenges of outsourcing. [5] (69 Ch)
18. Write short notes on: Out sourcing and off-shoring options. [2×4] (74 Ashwin)
19. Write short notes on: Outsourcing and off-shoring options. [4×5] (70 Ashad)